Adarsh Dwivedi

Applied ML and Deep Learning Engineer

+91-9305597756 | 23ucs509@lnmiit.ac.in | linkedin.com/in/adarshdwivedi30 | github.com/adarshcod30 | pypi.org/user/adarshcod30 | kaggle.com/adarshcod

Final-year CS engineer working across classical ML and deep learning, from graph-based fraud detection with a proved approximation bound to calibrated CNN ensembles with measured explanations. Built a platform that runs both together, LightGBM on station data fused with a CNN-LSTM over a 15,360-cell satellite grid. Placed 11th of 93 on a national leaderboard, ships to PyPI and Cloud Run, and publishes every limit beside the result.

EXPERIENCE

Deloitte India, Digital Excellence Centre **Jun 2026 - Aug 2026**

AI Engineer (Agentic AI) Intern Gurugram, India

- Migrated SAP Agentic, Deloitte's system that automates SAP configuration, from LangGraph running inside a FastAPI request handler onto a self-hosted Aegra agent server, with threads, runs and checkpoints in PostgreSQL inside the firewall at no licence cost.
- Proved the new path on the live SAP tenant: a Bank Accounting goal was planned into 8 configuration tasks, executed in a real browser through Playwright MCP, recovered from a duplicate-entry conflict and a mis-filled field, and finished 8 of 8 in about 15 minutes.
- Built capability in five layers on a multi-agent travel planner, from a single agent to a supervisor with three specialists, a human approval gate, real browser tools over MCP and serving through Aegra, proving each layer before starting the next.

PROJECTS

Orbweaver, Fraud Ring Detection in Transaction Graphs | Live | GitHub Python, igraph, Densest Subgraph, Knapsack

- Promotion abuse is a group behaviour, so scoring accounts one at a time misses it. Built a multi-relation account graph with edges weighted by entity rarity times measured fraud lift, fitted on training accounts only.
- Scored and pruned first, then extracted rings by greedy densest-subgraph peeling with a proved one-half approximation bound, reaching 0.7292 ring precision against a 0.2242 base rate on a 35.7M-edge graph.
- Attached the cost to every number: 0.371 genuine customers swept in per fraudster caught, an exact knapsack over reviewer minutes sizing the queue to analyst capacity, and 36 dated failures published.

OpenForensics, Deepfake Face Detection Ensemble | Live | GitHub PyTorch, ResNet50, VGG16, EfficientNetV2, Grad-CAM

- A detector that outputs only a label is unusable as evidence, so built a three-backbone ensemble of ResNet50, VGG16 and EfficientNetV2-B0 with calibrated confidence and an explicit borderline warning band.
- Trained on a two-stage schedule and served per-backbone Grad-CAM, so a reviewer sees which region drove each model's vote instead of trusting the aggregate score.
- Published the training, evaluation and robustness evidence behind every number in an eight-page dashboard, and stated the scope limits plainly: face crops only, swaps rather than generated images, and a single dataset.

Vayu, National Air-Quality Intelligence and Enforcement Platform *(Deployed on Google Cloud Run)* **| Live | GitHub** LightGBM, CNN-LSTM, DuckDB, Python

- Enforcement targets the dirtiest sensor, not the site where action pays most. Forecast air quality with LightGBM on station data plus a CNN-LSTM over a 15,360-cell satellite grid, then ranked interventions by Gaussian-plume modelled return.
- Kept PyTorch out of the API request path entirely, training the CNN-LSTM offline and serving from DuckDB as one embedded columnar file, so scoring is fast without a model server in the loop.

TECHNICAL SKILLS

Machine Learning: Scikit-learn, XGBoost, LightGBM, CatBoost, Optuna, Pandas, NumPy, Feature Engineering, Cross-Validation, Calibration, SHAP, A/B Testing, Statistics

Deep Learning: PyTorch, TensorFlow/Keras, CNNs, ResNet50, VGG16, EfficientNetV2, CNN-LSTM, Transformers, Transfer Learning, Ordinal Regression, Grad-CAM, OpenCV

Generative AI: LangGraph, LangChain, RAG, Prompt Engineering, Vector Search, Embeddings, Evaluation and Calibration, Amazon Bedrock, Gemini, OpenAI, Groq, Ollama

Backend and Deployment: FastAPI, Pydantic, Node.js, Express, REST APIs, PostgreSQL, MongoDB, DuckDB, Next.js, React, Streamlit

Languages: Python, C/C++, Java, JavaScript, TypeScript, SQL

ACHIEVEMENTS

- SWITCH Energy-X (India): 11th of 93 on the private leaderboard, recovering a hidden physical law from 500,000 unlabeled, 26%-missing sensor rows.
- KSP Datathon 2026 finalist: reached Level 2 with Kadi, a crime analytics platform built end to end over 50 days and deployed on Zoho Catalyst for the Karnataka State Police.

EDUCATION

The LNM Institute of Information Technology, Jaipur **2023 - 2027**

B.Tech, Computer Science and Engineering CGPA: 7.11 / 10

LEADERSHIP

- Coordinator, Debate Society, LNMIIT Jaipur (May 2025 - Apr 2026): led a 150+ member society across debates, discussions and inter-college competitions.
- Secretariat and Organizer, LNMIIT Model United Nations (Aug 2025 - Jan 2026): organised a large-scale MUN conference, coordinating delegates, committees, schedules and logistics.